Roomify

Evaluation Evidence

Detection quality and latency on held-out data, failure cases, behaviour without an AI key

Team Bootcamp2Roomify

Release candidate v0.9.0-rc1

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github.com/Bootcamp2Roomify/roomify_frontend · roomify_backend · roomify_cv

Evaluation Evidence

All numbers here were measured on 2026-10-24 by running the code in this repository; nothing is estimated or copied from model cards. Where something has **not been measured yet**, it says so. Reproduce with the scripts in `evaluation/scripts/`.

1. Furniture detection (CV service)

Set-up. 60 images from **COCO val2017** (held out from YOLO11n's training data, human box annotations): the 60 lowest image ids among the 334 val images that contain a bed or couch, i.e. real indoor rooms. 159 ground-truth objects in Roomify's six classes. Each image was sent through the product path — `POST /v1/analyze` on the running CV service (YOLO11n, `CONFIDENCE_THRESHOLD` 0.25) — and predictions were matched one-to-one per class at **IoU ≥ 0.5**; COCO crowd regions were ignored. "Large" = object covering ≥ 2 % of the image (the furniture a redesign is about).

Class	Ground truth	TP	FP	FN	Precision	Recall	Recall, large objects
bed	34	25	4	9	0.86	0.74	0.73 (33)
sofa	36	20	9	16	0.69	0.56	0.61 (33)
chair	37	22	12	15	0.65	0.59	0.68 (19)
table	11	6	9	5	0.40	0.55	0.67 (9)
tv	15	11	1	4	0.92	0.73	0.80 (10)
plant	26	15	7	11	0.68	0.58	0.67 (9)
all	**159**	**99**	**42**	**60**	**0.70**	**0.62**	**0.68 (113)**

Raw per-image results: `evaluation/data/detection_results.json`; image ids: `evaluation/data/coco_subset_ids.json`.

Latency (same 60 images, Apple-silicon laptop CPU, no GPU): model time p50 **32 ms**, p95 **41 ms**; HTTP round trip p50 35 ms, p95 44 ms, max 278 ms (first request, model warm-up).

What this means for the product. Roughly one object in three is missed and three in ten detections are wrong, so detection is a *starting list*, not an inventory — which is why the review step lets people rename, remove and add items, and why every detection under 60 % confidence is labelled "please check". Beds and TVs are reliable; tables are not (0.40 precision).

Failure cases

In each figure, green boxes are ground truth and red boxes are predictions with their confidence. Images are copyright their Flickr authors via COCO (annotations CC BY 4.0) and are shown for evaluation only.

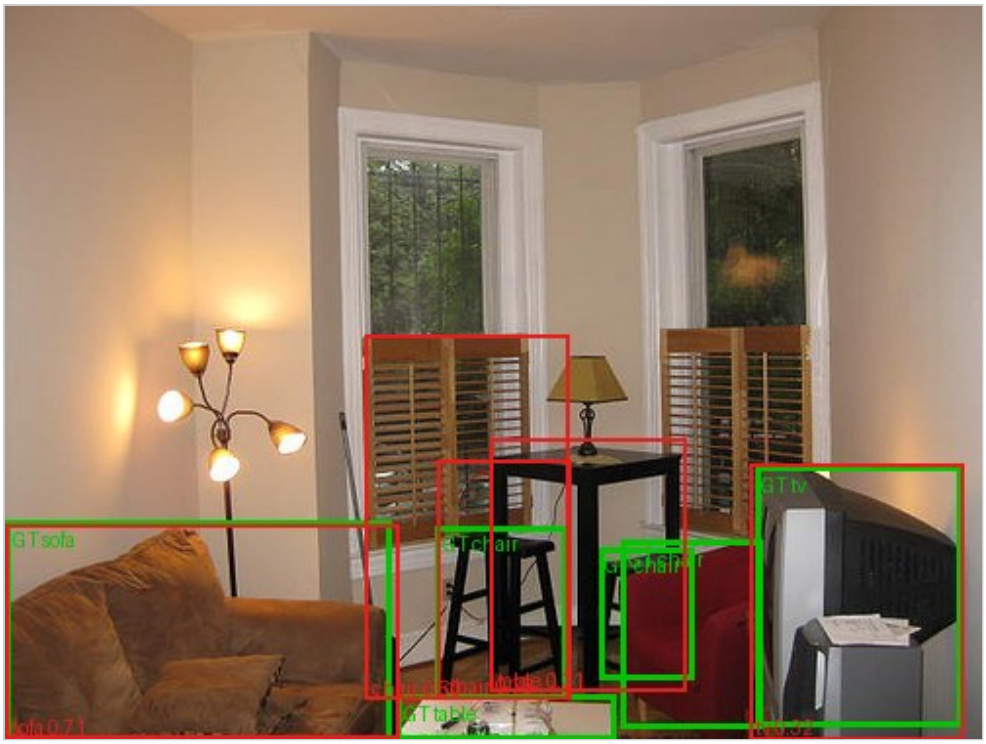


Figure 1. Failure case COCO 47769: a wooden window shutter is detected as a chair (confidence 0.50); two partly hidden chairs and a small table are missed; the floor lamp cannot be detected at all because COCO has no lamp class.



Figure 2. Failure case COCO 39477: the sofa and TV are found, but two hanging planters merge into one box (one false positive and one miss) and three small plants under 3 % of the frame are missed.



Figure 3. Failure case COCO 13923: in a bright living room the sofa (0.90) and large chairs are found, but seven smaller objects (each at most 5.5 % of the image) are missed; there are no false alarms.

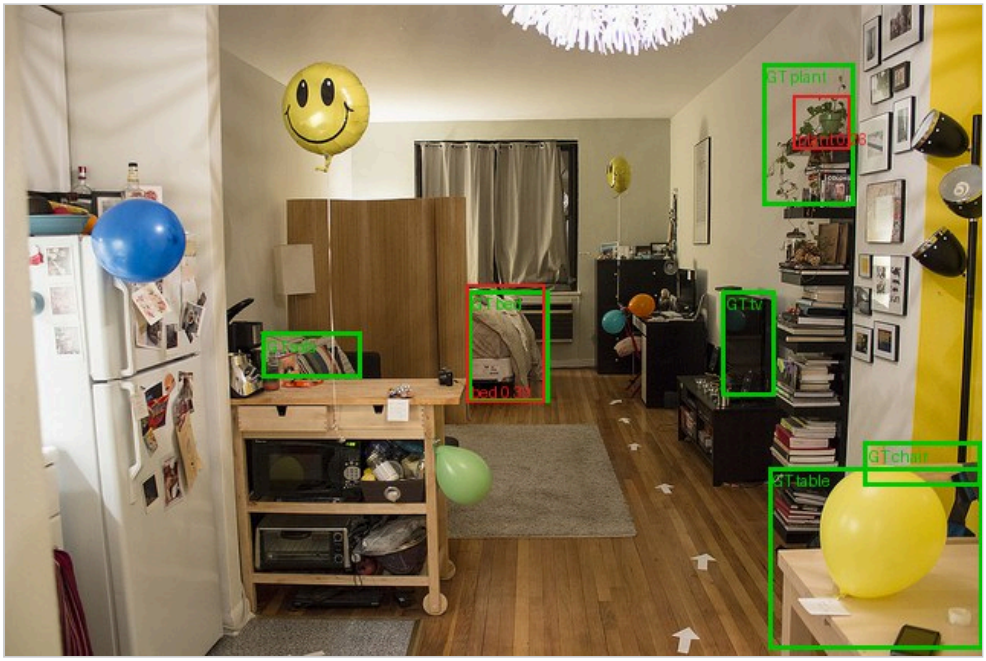


Figure 4. Failure case COCO 14038: in a cluttered studio the bed is found; the TV, a small table and other items under 5 % of the frame are missed. The shelf plant is detected but overlaps its annotation by less than IoU 0.5, so one localisation error counts as both a false alarm and a miss.

Limitations of this measurement

- 60 images is small: one more miss moves a class's recall by ~3 points. Confidence intervals are wide.
- COCO photos are not exactly what users upload (often wider, more cluttered, lower light).
- COCO has **no desk, wardrobe, shelf, lamp or rug classes**, so the most common student-room furniture is never detected; the scope's "storage furniture" relies entirely on manual entry.
- Measured at the service's single confidence threshold (0.25); no threshold tuning was done.

2. Redesign plan, visual concept and suggestions (OpenAI)

Pending: requires OpenAI key — no `OPENAI_API_KEY` has been configured, so none of these have been measured. No values are estimated or borrowed from model cards.

Result	Value
Plan budget compliance (first attempt / after corrective retry)	**Pending: requires OpenAI key**
Plan decision compliance (KEEP kept, REMOVE removed)	**Pending: requires OpenAI key**
Rental-mode compliance (no permanent work)	**Pending: requires OpenAI key**
Suggestion sets within budget; cheaper-alternative coverage	**Pending: requires OpenAI key**
DIY guides rejected by the safety filter	**Pending: requires OpenAI key**
Visual-concept quality (architecture preserved, kept items visible)	**Pending: requires OpenAI key**
Latency p50 / p95: plan, concept, suggestions, DIY	**Pending: requires OpenAI key**
Cost per full redesign (tokens × list price on the run date)	**Pending: requires OpenAI key**

The harness is ready:

- the backend logs one content-free line per provider call (`AI usage kind=text|image model=… promptTokens=… completionTokens=… reasoningTokens=… latencyMs=…`);
- `tests/live` in roomify_frontend runs the full journey with `LIVE_AI=1`.

Planned measurements (same rooms, fixed budgets of 150 / 300 / 600 USD, rental on and off):

Metric	How it will be measured
Plan budget compliance	share of first-attempt plans whose allocation sum equals the total and is ≤ budget (before the server's validator); share stored after the one corrective retry
Decision compliance	share of plans where every KEEP item is kept and every REMOVE item removed (validator outcome)
Rental compliance	share of rental-mode plans rejected for permanent-modification wording
Suggestion budget fit	share of first-attempt suggestion sets within budget; cheaper-alternative coverage
DIY safety	share of guides rejected by the safety filter
Latency	p50 / p95 per call type from the usage log
Cost	tokens × OpenAI list prices on the day of the run

3. Behaviour without an OpenAI key (verified)

`MissingAiKeyTest` runs the real OpenAI provider beans with no key; a blank key, an unedited `.env.example` placeholder and a key rejected by OpenAI (401/403) all count as "not configured".

Action	Response	State afterwards
Generate plan (twice)	`503 AI_CONFIGURATION_ERROR`	attempt stored as FAILED with that code, nothing pending, project status unchanged, retry allowed
Visual concept	`503 AI_CONFIGURATION_ERROR`	no image asset written
Suggestions	`503 AI_CONFIGURATION_ERROR`	no suggestion set written
DIY guide	`503 AI_CONFIGURATION_ERROR`	no guide written; saved plan still readable

The UI shows: "AI generation is not configured on this server yet (no valid OpenAI key). Your photo, furniture and preferences are saved — try again once AI is enabled." (frontend tests cover all four actions; the real-stack browser journey shows it for the plan).

4. Automated test results (2026-10-24, develop)

Suite	Result
Backend (JUnit, Postgres 17.11)	149 passed
CV service (pytest)	21 passed
Frontend unit (Vitest)	230 passed
Browser, stubbed API (Playwright, chromium + mobile)	39 passed, stable over 4 consecutive runs
Browser, fake Spring API through Next (Playwright)	6 passed
Browser, real stack — no AI key (Playwright `tests/live`)	journey, privacy, logout, CV-down manual entry: all passed